

Fair Cost Allocation in Energy Communities Under Forecast Uncertainty

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This work was supported by the German Research Foundation under Grant AL1185/9-1.

ABSTRACT Energy communities (ECs) are an increasingly studied path toward improving prosumer coordination. A central challenge of ECs is to allocate cost savings fairly to members. While many allocation mechanisms have been developed, existing literature does not account for the implications of inaccurate forecasts on the fairness of the allocation. We introduce a set of fairness conditions for imperfect knowledge allocation and show that these conditions constitute a Pareto front. We demonstrate how a well-established allocation scheme, the Shapley value mechanism (SVM), has unfavorable consequences for flexibility-providing community members and generally does not yield solutions on this Pareto front. In contrast, we interpret dispatch cost under imperfect knowledge as being composed of two components. The first represents the cost under perfect knowledge, and the second represents the cost arising from inaccurate forecasts. Our proposed mechanism extends an SVM-based allocation of the perfect knowledge cost by allocating the remaining cost in a way that guarantees finding solutions on the Pareto front. To this end, we formulate a convex multi-objective optimization problem that can efficiently be solved as a linear or quadratic program.

INDEX TERMS Energy community, cost allocation, fairness, forecast uncertainty, Shapley value, Pareto optimality.

I. INTRODUCTION

MODERN power systems face significant challenges due to the increasing adoption of intermittent renewable energy generation and complex power demand patterns from the rising electrification of heating and mobility. This increase in uncertainty leads to an increased economic risk for energy providers, which has to be mitigated by larger capital reserves. The opportunity cost of these reserves, in turn, raises operational expenditure, which is passed down to the end customers. This motivates targeting this challenge by increasing consumer coordination.

Demand-side management is based on providing consumers with incentives to adapt their demand with the goal of reducing overall energy costs and improving grid stability. While a vast range of such schemes exists [1], [2], [3], they face challenges in widespread adoption. According to surveys, consumers are particularly concerned about their own increased financial risk under dynamic pricing compared to the currently prevailing flat-rate tariffs [3].

Energy communities (ECs) alleviate this concern as their primary goal is to minimize energy cost for their members regardless of demand patterns. They are built around the paradigm that individual prosumers transparently share available energy excess with each other to avoid costly imports from outside the EC. It is well established that ECs save costs when compared to uncoordinated consumption [4], [5], [6], [7], [8]. Prosumers in EC are typically considered to have an inflexible power demand and to additionally own local renewable energy generation assets, or energy storage systems (ESS).

The central challenge for an EC is to design a cost (saving) allocation mechanism that is efficient and considered fair by all members. To this end, several proposals have been made, both from a perspective of non-cooperative [9] and cooperative game theory [10]. Cooperative approaches allocate cost ex-post based on some agreed algorithm, which has the central advantage of guaranteeing efficiency, i.e., the entire cost saving is allocated to the EC members.

Here, the concept of fairness is closely related to the principle of causation, i.e., each member pays the cost its participation caused. Under proportional allocation, costs are assigned proportionately to the consumption and production of each member [11], [12]. While such schemes can be designed to incentivize certain behaviors, they only indirectly relate to causation and require careful tuning.

In contrast, marginal-contribution-based mechanisms are explicitly based on causation. The approach that is regarded to best represent causation is the Shapley value mechanism (SVM) [13], which considers the marginal contribution of a member in every possible coalition. As its computation time is exponential in the number of EC members, several strategies have been developed to reduce computational effort [14], [15]. Applications in the EC context can be found in [16], [17], [18], with an overview given in [19].

An alternative approach from cooperative game theory is not designed around causation but rather around coalition stability. It is based on the nucleolus of the game and guarantees that the cost allocation is stable by minimizing the worst-case excess across all possible coalitions [5], [7], [8].

While the existing literature on cooperative cost allocation in ECs is highly sophisticated regarding game-theoretical robustness and computational efficiency, it does not define fairness in the presence of imperfect knowledge nor investigate the implications of inaccurate forecasts on individual members with different types of assets. Conversely, this paper explicitly explores cost allocation under imperfect knowledge with regard to several formal fairness conditions. Our specific contributions are

- a set of fairness conditions for imperfect knowledge allocation and the proof that these conditions constitute a Pareto front,
- a demonstration that the native application of the SVM under imperfect knowledge disadvantages members that provide dispatch flexibility and generally does not yield solutions on the Pareto front; and
- the formulation of a convex multi-objective optimal program that guarantees solutions on the Pareto front.

As the central focus is on fairness, we limit our study to the SVM, leaving the investigation of nucleolus-based approaches for future work. The remainder of this paper is organized as follows. After formalizing the investigated scenario, we introduce the required concepts from cooperative game theory and describe the SVM (Sec. II). We state the set of conditions our allocation must satisfy (Sec. III) and demonstrate that the SVM in the presence of imperfect knowledge does not satisfy the posed conditions (Sec. IV-A). Afterward, we proceed with the discussion of our proposed mechanism (Sec. IV-B). Finally, we evaluate our approach in two case studies (Sec. V) before drawing our conclusions (Sec. VI).

II. PRELIMINARIES

Since the specific system composition is irrelevant to the proposed allocation mechanism, we will focus on a high-level formalization of economic dispatch in ECs. Our EC consists of a set $\mathcal{N}$ of N members. Every member represents a residential or commercial consumer/prosumer and has one or several electrical components connected to it. The energy community is connected to the external grid M via a substation. The EC members delegate dispatch control to a central dispatch controller that controls components such as ESS $\mathcal{E}$, dispatchable generators $\mathcal{G}$, and trading with the external grid. Renewable generators $\mathcal{R}$ and inflexible loads $\mathcal{D}$ are considered disturbances. After a certain time, a billing period, the dispatch controller allocates the entire costs to the EC members. The dispatch controller is modeled as a self-contained algorithmic entity that the EC members democratically administrate.

A. SYSTEM MODEL

For conciseness, we present an intentionally simple system model that does not consider power flow constraints [5], [7], [8], [12], [18], [19], [20]. Including further variables and constraints would increase computational complexity, however, it would not impact the findings of this work.

1) INPUTS

The vector of control inputs $u_t = [p_t^{\mathcal{E}} \ p_t^{\mathcal{G}} \ p_t^M]^T$ at time t contains the active power injection p of ESS, generators, and the external grid. A positive power set point means consumption/charging and vice versa. The active power injections of inflexible loads and renewable generation are treated as uncontrollable exogenous inputs $d_t = [p_t^{\mathcal{D}} \ p_t^{\mathcal{R}}]^T$.

2) STATES

The state of the system x_t represents the state of charge of each storage system.

3) DYNAMICS

The state of charge of each ESS is subject to a difference equation $x_{t+1,i} = f_i(x_{t,i}, u_{t,i})$.

4) COST FUNCTION

When trading energy with the external grid, we assume a time-dependent convex market cost function $\psi_t(p_t^M)$ containing the origin [21, Sec. 10.1.15]. This implies that selling prices are never higher than buying prices and that the unit price for buying energy never decreases with increasing imports and the unit price for selling energy never increases with increasing exports. Thermal generators are commonly modeled with a quadratic cost function [22], [23]. We abstract this as a convex cost function $\omega_i(p_{t,i})$, which is non-zero only for $p_{t,i} \leq 0$, i.e., energy generation. As there is physical wear associated with charging/discharging processes, we model for every ESS a per-unit cost of wear $\rho_i \geq 0$ [5], [24]. Overall,

we write the operational cost of the EC for one time step as

$$c(u_t, \psi_t) = \psi_t(p_t^M) + \sum_{i \in \mathcal{G}} \omega_i(u_{t,i}) + \sum_{i \in \mathcal{E}} \rho_i |u_{t,i}|. \quad (1)$$

B. ECONOMIC DISPATCH

The optimal scheduling of power dispatch in the EC is a stochastic optimization problem as some quantities, such as renewable generation, power demand patterns, and market prices, are uncertain beforehand. Therefore, we consider them as random variables and assume forecasts for exogenous inputs $\hat{d}$, and market prices $\hat{\psi}$. More precisely, we denote the forecast of a value v at time t with the knowledge at time $t_0 \leq t$ as $\hat{v}_{t|t_0}$, where $\hat{v}_{t_0|t_0} = v_{t_0}$. Note that we are not limited to a certain model of forecast accuracy.

Economic dispatch is modeled as a discrete-time model predictive control problem as new information becomes available with time. We assume a receding horizon with access to forecasts for H time steps into the future and a known current state. In each control step t , the optimal dispatch problem considers a) the entire forecast horizon, b) that power injections sum to zero, and c) input and state limits:

$$\begin{aligned} u_{[t,t+H]}^* &= \underset{u_{[\cdot]}}{\operatorname{argmin}} \left(\sum_{\tau=t}^{t+H} c(u_\tau, \hat{\psi}_{\tau|t}) \right) \\ \text{s.t.} \quad & x_{\tau+1,i} = f_i(x_{\tau,i}, u_{\tau,i}) \quad \forall i \in \mathcal{E} \\ & 0 = \mathbf{1}^\top u_\tau + \mathbf{1}^\top \hat{d}_{\tau|t} \\ & u_\tau \in [\underline{u}_\tau, \bar{u}_\tau] \\ & x_\tau \in [\underline{x}_\tau, \bar{x}_\tau], \end{aligned} \quad (2)$$

where $\mathbf{1}$ is a vector of ones of appropriate size and $u_{[\cdot]}$ is the input trajectory for the entire horizon. Only the first entry u_t^* of the computed input trajectory is applied to the system. Each dispatch decision can either be solved considering only the expected values for generation, demand, and market prices as denoted here or explicitly considering their respective uncertainties as, for example, in [25]. While a probabilistic formulation can reduce the impact of inaccurate forecasts, our approach remains relevant as long as that impact is not entirely eliminated.

After a billing period of length H , the cost allocation algorithm is executed, requiring members to pay their allocated cost share. From now on, we will write the total cost of an EC composed of the set of nodes $\mathcal{N}$ resulting from solving (2) over one billing period as

$$C(\mathcal{N}) = \sum_{t=0}^H c(u_t^*, \psi_t). \quad (3)$$

Throughout this work, we will simply write C when considering the entire set of nodes $\mathcal{N}$. The existence of perfect forecasts is a special case of (2) for which $\forall \tau \geq t : \hat{d}_{\tau|t} = d_\tau, \hat{\psi}_{\tau|t} = \psi_\tau$. At the end of the billing period, we can simulate perfect knowledge ex-post by using the realized values as forecasts. In this case, we have to solve (2) only

once at $t = 0$ to obtain the perfect knowledge optimal input trajectory. We describe the cost that would have been incurred under this trajectory during the billing period as perfect knowledge cost $\tilde{C}$.

Note that (2) does not necessarily need to be an exact representation of the dispatch controller. The actual dispatch could be based on a highly complex nonlinear model, while a simplified model is used for the cost allocation. This would lead to an approximation of the Pareto front constituted by the fairness conditions introduced in Sec. III.

C. COST OF IGNORANCE

After one billing period, the EC has incurred the actual cost C based on dispatch decisions that took account of the forecasts available at the time and the actual later realizations of the forecasted quantities. Accordingly, we can view the final cost as being composed of the perfect knowledge cost and the cost of ignorance (CoI) $\check{C}$ arising from inaccurate forecasts¹:

$$C = \tilde{C} + \check{C}. \quad (4)$$

The CoI is related to the well-known concept of value of information [27, Ch. 15.6]. Specifically, it can be interpreted as the sum of the ex-post value of information for each dispatch decision over one billing period. By definition, the CoI is non-negative.

D. COALITIONAL GAME THEORY

Our allocation resorts to mechanisms developed in the field of coalitional game theory. We view $\mathcal{N}$ as the set of players and the grand coalition. Denoting a coalition of players as $\mathcal{S} \subseteq \mathcal{N}$, the cost for a specific coalition $\mathcal{S}$ is written as $C(\mathcal{S})$. The allocation scheme Φ calculates a cost share ϕ_i for each member $i \in \mathcal{N}$ as a mapping $\Phi(C) \rightarrow \phi$. We refer to ϕ_i as payment, although it might be negative. The payment under perfect knowledge is denoted by $\tilde{\phi}_i$.

We lastly introduce several concepts that are necessary for the discussion in the following sections.

Definition 1 (Marginal contribution). *The marginal contribution describes the cost difference of a coalition with and without the participation of member i and is defined as $C_{-i}(\mathcal{S}) = C(\mathcal{S} \cup i) - C(\mathcal{S})$.*

Definition 2 (Stand-alone cost). *We write a member i acting independently as a singular coalition $\mathcal{S}_i = \{i\}$. The corresponding stand-alone cost is defined as $C_i(\mathcal{S}_i) = \phi_i^a$. The stand-alone cost under perfect knowledge is written as $\tilde{\phi}_i^a$.*

Definition 3 (Regret). *The regret $\epsilon_i = \phi_i - \tilde{\phi}_i^a$ is given by the difference between payment and perfect knowledge stand-alone cost. The perfect knowledge regret is given by $\tilde{\epsilon}_i = \tilde{\phi}_i - \tilde{\phi}_i^a$.*

Note that members would realistically also suffer from inaccurate forecasts in stand-alone mode. However, we opt

¹This split is similar to the formulation of uncertain value functions in [26]. However, we attribute the uncertainty not to the members but to the dispatch controller.

for a guaranteed overestimation of the regret of a member by assuming perfect knowledge in stand-alone mode. If more information on forecasting models is available, we can adjust the definition of regret accordingly without impacting the validity of our approach.

E. SHAPLEY VALUE MECHANISM

The SVM is a well-established mechanism and is generally seen to be most closely related to the principle of fairness [10], [19]. It allocates cost based on the mean marginal contribution of each member over all possible coalitions $S \subseteq \mathcal{N}$. The Shapley value for a member i is defined as [13, Theorem1]

$$\phi_i^{\text{SVM}} = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!(N - |S| - 1)!}{N!} C_{-i}(S). \quad (5)$$

This formula can be interpreted as the sum of the marginal contribution of player i over all possible coalitions excluding player i , each weighted by the probability of player i being the marginal player in that coalition.

As the SVM scales exponentially in the number of participants, several approximations relying on random sampling of subcoalitions have been proposed [14], [15], [20], [28]. To maintain efficiency under sampling, a normalization factor is used [19].

III. PROBLEM STATEMENT

Fairness is a central condition for a communal cost allocation scheme. While fairness does not have a universal formal definition, the principle of causation can serve as an intuitive proxy. This can, however, only be defined for perfect knowledge forecasts since there is no clear causal link between the behavior of a member and the forecast accuracy of the dispatch controller. Therefore, we extend the notion of causation by comparing the cost caused by a member under perfect knowledge with its imperfect knowledge payment. We call this concept divergence. Additionally, we demand that the entire cost be allocated and that members have the incentive to join the EC. While we cannot guarantee that the regret of all members is negative, we aim to minimize the maximum regret across the EC.

In summary, we pose the following conditions to our allocation mechanism:

Condition 1 (Causation). *Under perfect knowledge, the payment of each member directly depends on the cost they caused through their participation. Causation is represented by the conditions of symmetry, linearity, and null player [29]:*

- *Symmetry:* $\forall S \subseteq \mathcal{N} \setminus \{i, j\}: \tilde{C}(S \cup \{i\}) = \tilde{C}(S \cup \{j\}) \rightarrow \tilde{\phi}_i = \tilde{\phi}_j$.
- *Linearity:* For two costs C, C' and $\alpha \in \mathbb{R}$ we have $\forall i \in \mathcal{N}: \alpha \tilde{\phi}_i(C) + \alpha \tilde{\phi}_i(C') = \tilde{\phi}_i(\alpha C + \alpha C')$.
- *Null player:* $\forall S \subseteq \mathcal{N} \setminus \{i\}: \tilde{C}(S \cup \{i\}) = \tilde{C}(S) \rightarrow \tilde{\phi}_i = 0$.

Condition 2 (Efficiency). *The entire EC cost is allocated to its members [29]: $\sum_i \phi_i = C$.*

Condition 3 (Minimum divergence). *The deviation from the perfect knowledge allocation is minimized. This is interpreted as the minimization of the p -norm*

$$\min_{\phi} \|\phi - \tilde{\phi}\|_p, \quad (6)$$

where $p \geq 2$.

Condition 4 (Minimum worst-case regret). *The maximum, i.e., worst-case, regret across all EC members is minimized:*

$$\min_i \max_{\epsilon_i} (\phi_i - \tilde{\phi}_i^a). \quad (7)$$

Lemma 1. *The conditions 1 – 4 cannot be concurrently fulfilled in the general case and, therefore, constitute a Pareto front.*

Proof: Minimum divergence as defined in (6) is achieved if all members pay an equal share of the CoI in addition to their perfect knowledge payment. The only case in which this equal split represents the minimization of worst-case regret from (7) is if all members have an identical perfect knowledge regret. This is generally not true. Therefore, any improvement in worst-case regret increases divergence and vice versa, constituting a Pareto front. $\square$

Accordingly, our allocation must find solutions on the Pareto front constituted by the conditions 1 – 4.

IV. COST ALLOCATION

A. SHAPLEY VALUE MECHANISM UNDER IMPERFECT KNOWLEDGE

It seems a natural choice to simply apply the SVM to the imperfect knowledge cost. However, the SVM generally does not yield solutions on the Pareto front constituted by conditions 1 – 4. This is due to the fact that the SVM allocates most of the CoI to the subset of members that provide dispatch flexibility. We illustrate this effect in the following toy example that is additionally visualized in Fig. 1.

Example 1. *The example system consists of one member with a constant inflexible demand of 1.0kW and one member with an ESS with a power limit of 1.0kW and a maximum state of charge of 1.0kWh. We consider 4 time steps with a sample time of 1 hour. At time step 1, the market price is at 1.0€/kWh. The dispatch controller expects the price to rise to 2.0€/kWh between time steps 2-3 and afterward reduce to 0.5€/kWh. However, the forecasts are wrong and the price reduces to 0.0€/kWh between time steps 2-3 and afterward increases to 0.5€/kWh (see Fig. 1a). If no ESS were available, the dispatch controller would have to satisfy demand at the current price, leading to a total cost of 1.5€ (see Fig. 1b). If dispatch controller perfectly predicted the actual price trajectory, it would choose to charge the ESS between time steps 2-3 and use that energy between time steps 3-4, resulting in a total cost of 1.0€ (see Fig. 1c). However, based on its inaccurate forecasts, the dispatch controller expects a cost of 3.5€ without dispatching the ESS. Accordingly, it plans to charge the ESS between steps 1-2*

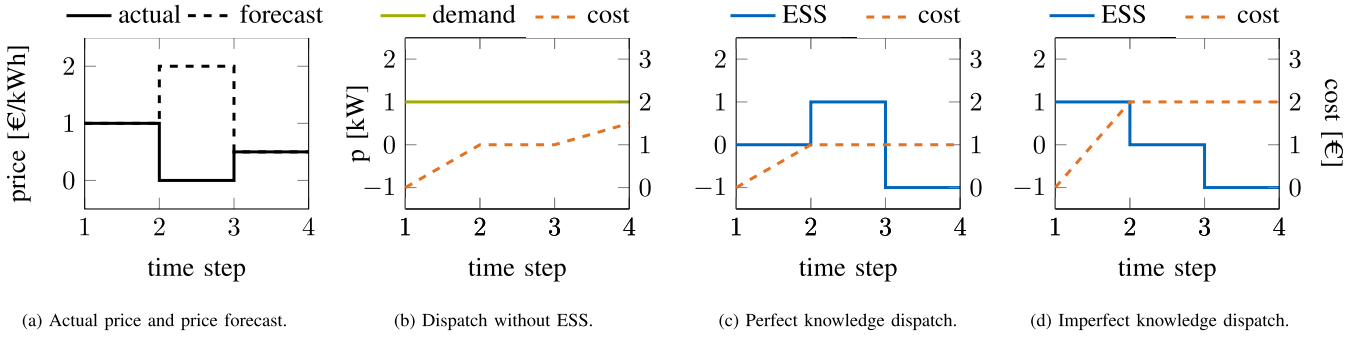


FIGURE 1. Scenario from example 1 illustrating how the availability of an ESS can increase cost under imperfect knowledge.

and use the stored energy between time steps 2-3 with a projected total cost of 2.5€. However, since the price reduces at time step 2, the dispatch controller imports energy between time steps 2-3 and uses the stored energy between time steps 3-4, leading to a realized cost of 2.0€ (see Fig. 1d). Overall, the realized cost with ESS is higher than it would have been without any dispatch flexibility. Even though the member providing storage capacity was not responsible for the inaccurate forecasts of the dispatch controller, the relatively higher cost would be attributed to it under the SVM since its participation enabled the costly decision of storing energy for later.

Theorem 1. *The SVM does not guarantee solutions on the Pareto front constituted by conditions 1 – 4.*

Proof: Example 1 can serve as an instance for which the allocation of the SVM is not on the Pareto front. We define the member with ESS as player i and the member with the inflexible load as player j and omit the currency unit in the following argument. Assuming that energy can be sold at the current market price, we can compute $\tilde{\phi}_i^a = -0.5$, $\tilde{\phi}_j^a = 1.5$, $\tilde{\phi}_i = -0.5$, $\tilde{\phi}_j = 1.5$. The SVM allocation under imperfect knowledge yields $\phi_i = 0.5$, $\phi_j = 1.5$, allocating the CoI of 1.0 solely to player i . Accordingly, the regrets are $\epsilon_i = 1.0$, $\epsilon_j = 0.0$ and the divergence is $\|\phi - \tilde{\phi}\|_2 = 1.0$. For this scenario, the Pareto front collapses into one point, since both members have an identical perfect knowledge regret of 0.0. This point is obtained by evenly splitting the CoI, resulting in the payments $\phi_i = 0.0$, $\phi_j = 2.0$, a regret of 0.5 for both members, and a divergence of approximately 0.71. It is easy to see that the SVM yields a solution that has both higher divergence and higher worst-case regret than the Pareto optimal point, which concludes the proof. $\square$

Remark 1. *The SVM yields solutions that are often far away from the Pareto front because the SVM generally neither minimizes divergence nor minimizes worst-case regret and is, in some aspects, in direct opposition. As any arising CoI is included in the Shapley value of all coalition members with the same weight, the CoI share allocated to a member with ESS can never be lower than the CoI share allocated to a member without ESS, and is usually larger. Since a more unequal allocation of the CoI leads to higher divergence, the*

solutions of the SVM are generally removed from the Pareto front in this dimension. To minimize worst-case regret, the allocation would need to assign a lower payment to a member with higher perfect knowledge stand-alone cost: $\tilde{\phi}_i^a > \tilde{\phi}_j^a \rightarrow \phi_i < \phi_j$. Since the SVM is agnostic with respect to the perfect knowledge stand-alone cost, any allocation fulfilling this condition would be purely coincidental.

An increase in local generation often increases the utilization of storage systems, which leads to an increased allocation of CoI to members with local generation. This is especially pronounced when selling prices are much lower than buying prices, limiting the potential of arbitrage trading. In summary, the direct application of the SVM under imperfect knowledge penalizes the members who are most valuable to the community.

Note that it is difficult to link forecast accuracy to total cost straightforwardly. Rather, the deciding factor is if the inaccuracy leads to different dispatch decisions and to what extent. In the context of Example 1, the price forecast for the time between time steps 2-3 could be far higher than 2.0€/kWh without changing the imperfect knowledge dispatch. In contrast, if the forecast were a price of -2.0€/kWh, i.e., an equal absolute error but in the opposite direction, the imperfect knowledge dispatch would be identical to the perfect knowledge dispatch. However, we can generally expect forecast errors to be positively correlated to the CoI and to the mismatch between the native allocation and Pareto optimal points. We conduct a numerical investigation in Sec. V-A that validates this hypothesis.

B. A FAIR IMPERFECT KNOWLEDGE ALLOCATION

In contrast to the SVM, we take motivation from Lemma 1, and base our allocation on a multi-objective optimal program. First, we structure the payment into two elements:

$$\phi = \tilde{\phi} + \gamma \tilde{C}, \quad (8)$$

where $\gamma \geq 0$. Here, $\tilde{\phi}$ is the share of the perfect knowledge cost computed with the SVM,² and γ is the share of the CoI. This recasts the task of our allocation to finding γ .

²As mentioned in the introduction, the focus of this work is on fairness as opposed to coalition stability. However, it should be noted that while the SVM cannot guarantee stability in a context such as ours [7], [16], previous numerical analysis has shown that the stability benefit of nucleolus-based approaches can be rather small [5], [7].

Lemma 2. Under the payment structure from (8), we can represent the conditions 1–4 in the multi-objective optimization problem

$$\min_{\gamma, z} \{ \|\gamma \check{C}\|_p, z \} \quad (9a)$$

s.t.

$$1 = \sum_i \gamma_i \quad (9b)$$

$$0 \leq \gamma_i$$

$$z \geq \tilde{\epsilon}_i + \gamma_i \check{C},$$

where z is a real-valued scalar.

Proof: We address each condition separately.

1) CAUSATION

Under perfect knowledge, the CoI is zero, and our allocation is identical to the SVM, which fulfills symmetry, linearity, and null player [29]. Accordingly, the condition of causation is satisfied.

2) EFFICIENCY

The condition of efficiency is given by $C = \sum_i \phi_i = \sum_i \tilde{\phi}_i + \sum_i \gamma_i \check{C}$. Leveraging the definition of C from (4), we have $C = \tilde{C} + \check{C} = \sum_i \tilde{\phi}_i + \check{C}$ and can rewrite the condition of efficiency as $\sum_i \gamma_i = 1$.

3) MINIMUM DIVERGENCE

We represent the condition of minimum divergence in the cost function. With the payment structure from (8), we obtain $\|\phi - \tilde{\phi}\|_p = \|\tilde{\phi} + \gamma \check{C} - \tilde{\phi}\|_p = \|\gamma \check{C}\|_p$.

4) MINIMUM WORST-CASE REGRET

We transform the min-max of ϵ into the minimization of a scalar upper bound z , which allows us to include z in the objective function [30, Eq. 1.4]. We constrain z to be an upper bound for the regret with $z \geq \epsilon_i = \phi_i - \tilde{\phi}_i^a = (\tilde{\phi}_i + \gamma_i \check{C}) - \tilde{\phi}_i^a = \tilde{\epsilon}_i + \gamma_i \check{C}$. $\square$

There exist a multitude of approaches for solving multi-objective optimal programs [31]. We follow a well-established method that combines all objective functions in a weighted sum, which has been shown to be a sufficient condition for Pareto optimality [31], [32], [33]. For this method, it is standard practice to normalize the individual objective functions to a range of [0, 1], improving the interpretability of the objective function weights. A generic objective function $J_i(x)$ is normalized by [31, Eq. 7]

$$J'_i(x) = \frac{J_i(x) - \underline{J}_i}{\bar{J}_i - \underline{J}_i}. \quad (10)$$

Here, $\underline{J}_i$ describes the lower bound of $J_i(x)$ that is computed by exclusively minimizing this objective function subject to the original constraints. Conversely, $\bar{J}_i$ describes the upper bound of $J_i(x)$ that is computed by $\max_{j \in \mathcal{J}} J_i(x_j^*)$, where x_j^* is the point that minimizes objective function j and $\mathcal{J}$ is the set

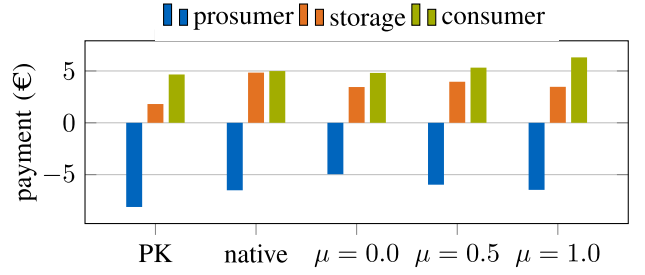


FIGURE 2. Absolute payment per household under perfect knowledge (PK), native SVM (native), and the proposed allocation with $p = 2$ and different values for μ .

of objective functions. Following this approach, we provide a concrete implementation of (9) in the following theorem.

Theorem 2. We are guaranteed to find solutions on the Pareto front constituted by the conditions 1 – 4 by solving the multi-objective optimal program

$$\begin{aligned} \min_{\gamma, z} & (1 - \mu)J^R(z) + \mu J^D(\gamma) \\ \text{s.t.} & (9b) \end{aligned} \quad (11)$$

$$J^R(z) = \frac{z - z^{R*}}{\bar{\epsilon} + \frac{\check{C}}{N} - z^{R*}}$$

$$J^D(\gamma) = \frac{\|\gamma\|_p - N^{\frac{1}{p}-1}}{\|\gamma^{R*}\|_p - N^{\frac{1}{p}-1}},$$

where $\mu \in [0, 1]$ is the objective function weight, z^{R*} is the objective function value with the associated CoI shares γ^{R*} obtained from solving (9) exclusively considering the minimization of z , and $\bar{\epsilon} = \max_i \tilde{\epsilon}_i$ is the maximum perfect knowledge regret among the members.

Proof: We first show that $J^R(z)$ and $J^D(\gamma)$ are valid normalizations for the two objective functions in (9a) based on the normalization from (10). Afterward, we discuss Pareto optimality.

1) NORMALIZED WORST-CASE REGRET

We obtain the lower bound for worst-case regret by solving (9) exclusively considering the minimization of z , which yields z^{R*} . The upper bound is reached when minimizing divergence exclusively. Here, the CoI would be evenly split, which results in a maximum worst-case regret of $\bar{\epsilon} + \frac{\check{C}}{N}$.

2) NORMALIZED MINIMUM DIVERGENCE

The general expression for minimum divergence is given by $\|\gamma \check{C}\|_p = (\sum_{i \in \mathcal{N}} |\gamma_i \check{C}|^p)^{\frac{1}{p}}$. Straightforwardly, the lower bound of this is reached when the CoI is split evenly, i.e., $\forall i: \gamma_i = \frac{1}{N}$, resulting in $(\sum_{i \in \mathcal{N}} |\frac{1}{N} \check{C}|^p)^{\frac{1}{p}} = (\frac{\check{C}^p}{N^{p-1}})^{\frac{1}{p}} = N^{\frac{1}{p}-1} \check{C}$. The upper bound can be computed by solving (9) exclusively considering the minimization of z , yielding $\|\gamma^{R*} \check{C}\|_p = \|\gamma^{R*}\|_p \check{C}$. Since $\check{C}$ appears in each term of the resulting objective function, it can be omitted.

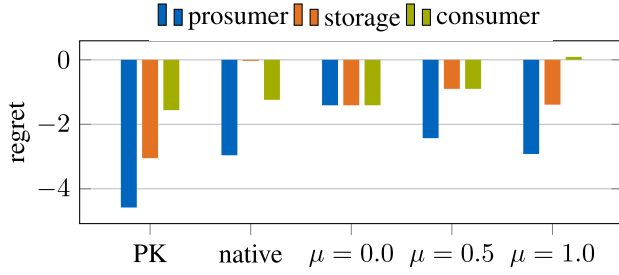


FIGURE 3. Absolute regret per household under perfect knowledge (PK), native SVM (native), and the proposed allocation with $p = 2$ and different values for μ .

3) PARETO OPTIMALITY

The optimization problem in (11) is a special case of the optimal program from Lemma 2 for which we have already shown that the conditions 1 – 4 are represented. Since the weighted sum approach is a sufficient condition for Pareto optimality [31], [32], [33], we are guaranteed to find points on the Pareto front constituted by these conditions. $\square$

The choice of p -norm determines the computational efficiency of the optimal program (11). In the following Corollary, we provide a more detailed statement for some natural choices of p .

Corollary 1. *The optimal program (11) is convex. Specifically, for $p = \infty$, we need to solve two linear programs. For $p = 2$, we need to solve one linear and one quadratic program.*

Proof: Convexity follows readily from the fact that we have a linear combination of convex cost functions with solely linear constraints. To obtain the normalized objective functions, we first need to solve (9) exclusively considering the minimization of z , which is a linear program independent of p .

For $p = \infty$, we minimize the maximum CoI share with $\|\gamma\|_\infty$. Accordingly, we can replace $\|\gamma\|_\infty$ with a scalar y and enforce y to be an upper bound by adding the constraint $y \geq \gamma_i$. Furthermore, we have $\lim_{p \rightarrow \infty} N^{\frac{1}{p}-1} = \frac{1}{N}$. Overall, we obtain a linear program.

For $p = 2$, $J^D(\gamma)$ contains square roots. However, we can obtain a quadratic program by replacing $J^D(\gamma)$ with its square

$$J^D(\gamma)^2 = \frac{(\sum_i \gamma_i^2) - N^{-1}}{(\sum_i (\gamma_i^{R*})^2) - N^{-1}}.$$

Since $\text{argmin}_\gamma J^D(\gamma) = \text{argmin}_\gamma J^D(\gamma)^2$, we are still guaranteed to find points on the Pareto front. $\square$

V. CASE STUDIES

The first case study is a simple system of three households tailored to demonstrate the implications of different imperfect knowledge allocations on different types of EC members. The second case study is based on a realistically sized rural EC and evaluated under coalition sampling. We utilize the

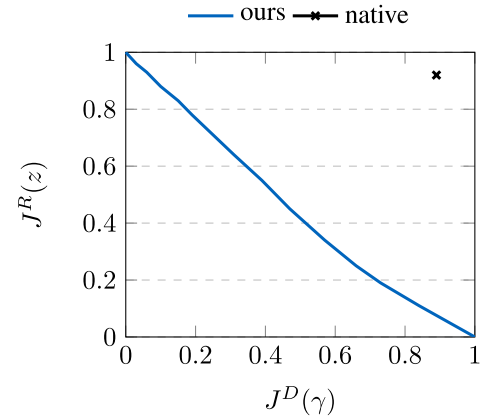


FIGURE 4. Visualization of the Pareto front for the three household scenario with $p = 2$. The x mark signifies the value of the native SVM allocation.

TABLE 1. Pearson correlation coefficients between CoI, volume-weighted mean absolute forecast error (MAE), and minimum Euclidean distance of the native allocation to the Pareto front (Dist) in the three household scenario.

	CoI	MAE	Dist
CoI	1.00	0.64	0.52
MAE	0.64	1.00	0.63
Dist	0.52	0.63	1.00

open-source Python library *CommonPower* [34] to model and simulate our scenarios. All generation and load profiles are based on the SimBench database [35], namely the *1-LV-rural2-1-sw* dataset.³ The forecasts for all quantities are their values at the same time of day one week prior unless otherwise mentioned. We model ESS dynamics as the piecewise linear function

$$f_i(x_{t,i}, u_{t,i}) = \eta_i^S x_{t,i} + \begin{cases} \eta_i^C u_{t,i} & \text{if } u_{t,i} \geq 0 \\ \frac{1}{\eta_i^D} u_{t,i} & \text{if } u_{t,i} < 0, \end{cases} \quad (12)$$

where $\eta_i^S \in [0, 1]$ is the self-discharge rate, $\eta_i^C \in [0, 1]$ is the charge efficiency, and $\eta_i^D \in [0, 1]$ is the discharge efficiency of ESS i .

A. THREE HOUSEHOLDS

The first case study involves three stereotypical households. All households have identical inflexible demand. The first household is a prosumer and additionally owns a photovoltaic (PV) generator. The second household is a storage provider, additionally owning an ESS. The third household is a pure consumer with no additional devices. The buying price is constant at 0.34€/kWh and the selling price at 0.08€/kWh. We assume forecasts for load and generation as described above and perfect knowledge of the prices. We start the simulation on 17.03.2016 at 00:00 with both the forecast horizon and the billing period set to 72 hours. In this setting, the total

³<https://simbench.de/de/datensaetze>

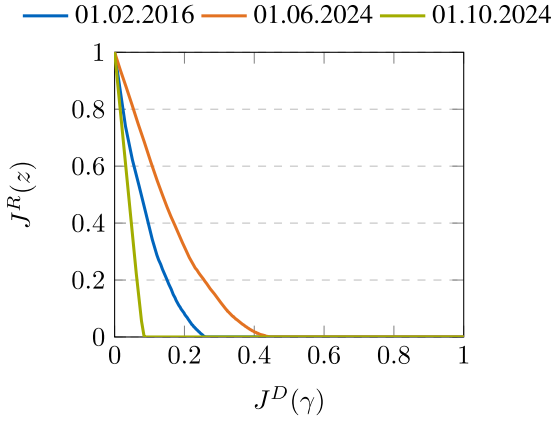


FIGURE 5. Visualization of the average Pareto fronts with $p = 2$ across five random system realizations for each test day on the 27-bus Kerber network.

load of all households over the billing period is 214.97kWh, the total PV generation is 88.89kWh, and the ESS capacity is 10kWh.

Fig. 2 shows the absolute payment, and Fig. 3 shows the absolute regret per household for $p = 2$ and different values for the weight parameter μ from Theorem 2. The CoI in this scenario is 4.96€. It is evident that the native approach allocates the majority of the CoI to the prosumer (32.53%) and storage provider (60.90%), while the payment of the consumer household rises only by a small amount (6.57%). This skewness was explained in Example 1 and is the central motivation of this work. The allocations based on the proposed approach differ depending on the choice of the weight μ . We can see the expected pattern that with rising μ the payment distribution more closely resembles the perfect knowledge allocation at the price of ever-increasing worst-case regret. For $\mu = 0$, we solely minimize the worst-case regret, which leads to the CoI being almost entirely allocated to the prosumer and storage provider to increase the regret of the consumer only as little as possible. Accordingly, 63.86% of the CoI is allocated to the prosumer due to its large negative perfect knowledge regret. Conversely, for $\mu = 1$, we solely minimize the maximum deviation from the perfect knowledge allocation. The CoI is split evenly, increasing the regret for all households by an identical amount. In this case, this leads to a positive regret for the consumer household. When both objectives are taken into account with $\mu = 0.5$, a middle path is chosen for which the maximum deviation is limited with a CoI share of 43.22% while all households maintain a comfortably negative regret.

We can approximately visualize the Pareto front by uniformly sampling the weight coefficient μ . For the three household scenario, the result is provided in Fig. 4 with $p = 2$. Here, the trade-off between the two objectives becomes apparent. The corresponding normalized objective function value of the native allocation is included for reference. It is significantly worse than any point on the Pareto front.

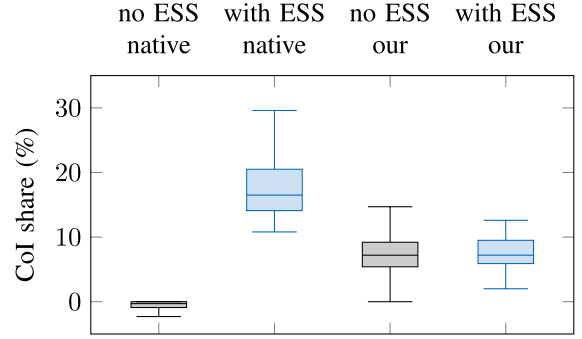


FIGURE 6. Boxplot of the CoI shares of members with an ESS (29 samples) and members without an ESS (41 samples) in the 27-bus Kerber network across five random system realizations on 01.06.2016. We compare the native SVM (native) with the proposed approach (our) using $p = 2$, $\mu = 10^{-8}$. The whiskers contain all data points.

For a numerical investigation on the impact of forecast accuracy, we simulate the scenario ten times, each with forecasts of PV generation and household loads as the values 1-10 weeks prior, respectively. We compute the mean average error across the 72-hour horizon and calculate a volume-weighted average of generation and load forecast error (MAE). Additionally, we determine the minimum Euclidean distance (Dist) of the native allocation to the Pareto front in the coordinates of the normalized objective functions. We report the Pearson correlation coefficients between CoI, MAE, and Dist in Table 1. We can observe a moderately strong positive correlation between all considered metrics. This provides evidence for our hypothesis that the CoI tends to increase with increasing forecast error and that the native allocation tends to be further removed from the Pareto front with increasing forecast error.

B. 27-BUS KERBER NETWORK

In the second experiment, we demonstrate the properties of our proposed approach on a realistically sized rural network and under coalition sampling. The topology of the test system is based on the *Kerber Landnetz Kabel 2* network in the *pandapower* library [36], representing an average rural network based on a theoretical power network categorization [37]. The network comprises 30 buses, with 14 designated as households, and is connected to the external grid through a substation. For the simulation, components are randomly assigned to households. Each household is equipped with an inflexible load (100% probability), a battery (50% probability), and a PV generator (75% probability). All component parameters are selected independently and uniformly from specified ranges. We assume a linear market cost function and use spot market prices from Denmark.⁴ The forecast horizon and billing period are 72 hours. We simulate five different random system realizations for three different days in 2016

⁴<https://www.energidataservice.dk/tso-electricity/Elspotprices>

TABLE 2. Comparison of the native SVM (native) and the proposed allocation (ours) with $p = 2$, $\mu = 0.5$ in the 27-bus Kerber network. The first value is the average, and the value in brackets is the minimum across the five random system realizations. We also report the total load, PV generation, and ESS capacity over the 72-hour simulation period averaged across all system realizations.

Date	01.02.2016		01.06.2016		01.10.2016	
Method	native	ours	native	ours	native	ours
CoI [€]	3.28		8.02		1.57	
J^D	0.98 (0.59)	0.25 (0.08)	1.43 (0.74)	0.35 (0.20)	0.41 (0.32)	0.08 (0.08)
J^R	1.33 (0.04)	0.01 (0.00)	1.66 (0.85)	0.06 (0.00)	0.76 (0.01)	0.00 (0.00)
Total load [kWh]	1610.55		662.72		661.18	
Total PV generation [kWh]	549.45		819.80		763.71	
Total ESS capacity [kWh]	80.68		82.20		81.33	

(01.02., 01.06, 01.10.) and approximate the Shapley value with the approach from [14].

Table 2 depicts the average and minimal normalized cost function values across all test days for the native SVM and our approach with $p = 2$ and $\mu = 0.5$. Both the average and the minimum values are consistently lower with our approach. Since we know that each sample of our allocation is a Pareto optimal point, we can state that the native SVM yields allocations that are generally far removed from the Pareto front, a result in line with Fig. 4. Furthermore, we observe that the CoI varies significantly between the different dates. This is largely due to the specific realizations for each simulation period. However, we can draw a general connection between the ratio of load and generated energy within the community. This generation-to-load ratio influences the saving potential of the EC and, accordingly, the potential for erroneous planning. This leads to the CoI being larger when more energy is generated relative to power demand.

We find that the cost function values are positively correlated with the CoI, which corresponds to Fig. 5 depicting the point-wise average Pareto fronts across the five system realizations for the three test days. In addition to higher cost function values, the trade-off between the two objectives is more pronounced when the CoI is higher. In correspondence with the insights from Figs. 2 and 3, this can be explained as follows: Starting at the top left of the Pareto plot, we allocate an equal share of the CoI to every member. Increasing the parameter μ , we allocate less CoI to the member with the highest perfect knowledge regret. At a certain point, all members have equal regret which is the lower bound for the minimization of worst-case regret, i.e., $J^R(z) = 0$. The higher the generation-to-load ratio, the lower the perfect knowledge regret of prosumers and storage providers in comparison to consumers. This means reaching the lower bound of worst-case regret requires the CoI to be allocated more unequally, which pushes the solution further away from the perfect knowledge allocation. This leads to higher divergence, i.e., a larger value for $J^D(\gamma)$.

Lastly, we visualize the difference in allocated CoI share between members with an ESS and members without an ESS for 01.06.2016 in Fig. 6. The boxplot shows the substantial disadvantageous imbalance against flexibility-providing members. In fact, for this set of samples, the members owning an ESS are allocated more than the entire CoI, while the other

members generally pay less than their perfect knowledge payments. In contrast, the proposed mechanism does not exhibit this behavior, with the allocations of both groups being largely in balance. For an expressive comparison, we use $\mu = 10^{-8}$ in our approach, which puts a large weight on the minimization of worst-case regret and leads to more diverse CoI shares.

Both experiments support our theoretical insight that the native application of the SVM under imperfect knowledge disadvantages flexibility-providing members. While the proposed allocation resolves this imbalance, the trade-off between the two objectives of minimum divergence and minimum worst-case regret becomes apparent. The Pareto front is generally highly non-symmetric, which can make the choice of the weighting parameter μ challenging for practitioners. Considering that the solutions of the native SVM are generally far removed from the Pareto front, there is strong justification for basing the allocation on the proposed multi-objective optimization. However, further research would be beneficial for navigating the central trade-off.

VI. CONCLUSION

This paper proposes an approach for fair cost allocation in energy communities under imperfect knowledge. We present a structured approach to allocate the cost of erroneous planning, the cost of ignorance, in a fair way. To this end, we formulate a multi-objective optimization problem that guarantees finding solutions on the Pareto front constituted by a set of conditions. This optimal program is convex and can for natural parameter choices be efficiently solved as a linear or quadratic program. Furthermore, we show that a native application of the popular Shapley value mechanism under imperfect knowledge disadvantages members that are providing flexibility. Our results from two case studies confirm that the allocations of the native SVM are generally far removed from the Pareto front of the posed conditions and that the proposed approach does not disadvantage flexibility-providing members.

ACKNOWLEDGMENT

The authors would like to thank Adrian Kulmburg for many helpful discussions.

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